

MATHEMATICS BY D.K. SINGH

Topic: Matrices and Determinants Test COMPLETE SOLUTION BOOKLET & ANSWER KEY

PART 1: ANSWER KEY

Section A: Multiple Choice Questions

Q.	Ans.	Q.	Ans.	Q.	Ans.	Q.	Ans.	Q.	Ans.
1	(b)	7	(b)	13	(b)	19	(c)	25	(b)
2	(b)	8	(c)	14	(c)	20	(d)	26	(d)
3	(b)	9	(c)	15	(a)	21	(b)	27	(a)
4	(a)	10	(a)	16	(b)	22	(c)	28	(b)
5	(d)	11	(c)	17	(b)	23	(a)	29	(a)
6	(a)	12	(b)	18	(c)	24	(a)	30	(b)

Section B: Integer Type Questions

Q.	Ans.	Q.	Ans.	Q.	Ans.	Q.	Ans.	Q.	Ans.
31	1	33	0	35	4	37	10	39	8
32	16	34	0	36	3	38	9	40	6

PART 2: DETAILED STEP-BY-STEP SOLUTIONS

SECTION A: MULTIPLE CHOICE QUESTIONS

1. **Element a_{12} calculation:**

Given $a_{ij} = \frac{(i+2j)^2}{2}$. For a_{12} , substitute $i = 1$ and $j = 2$:

$$a_{12} = \frac{(1+2(2))^2}{2} = \frac{(1+4)^2}{2} = \frac{25}{2}$$

Correct Option: (b)

2. **Properties of $A - A^T$:**

Let $B = A - A^T$. Taking transpose on both sides:

$$B^T = (A - A^T)^T = A^T - (A^T)^T = A^T - A = -(A - A^T) = -B$$

Since $B^T = -B$, the matrix is always a Skew-symmetric matrix.

Correct Option: (b)

3. **Determinant value of $|\text{adj } A|$:**

Using property $|\text{adj } A| = |A|^{n-1}$ where $n = 3$ and $|A| = 5$:

$$|\text{adj } A| = 5^{3-1} = 5^2 = 25$$

Correct Option: (b)

4. **Computing $A^2 - 5A$ for $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$:**

First compute A^2 :

$$A^2 = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 1+6 & 2+8 \\ 3+12 & 6+16 \end{bmatrix} = \begin{bmatrix} 7 & 10 \\ 15 & 22 \end{bmatrix}$$

Now compute $A^2 - 5A$:

$$A^2 - 5A = \begin{bmatrix} 7 & 10 \\ 15 & 22 \end{bmatrix} - \begin{bmatrix} 5 & 10 \\ 15 & 20 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = 2I$$

Correct Option: (a)

5. **Total number of possible matrices:**

A 3×3 matrix contains 9 elements. Since each element has 2 options (0 or 1), the total number of configurations is:

$$2^9 = 512$$

Correct Option: (d)

6. **Nature of $AB - BA$:**

Given $A^T = A$ and $B^T = B$. Let $C = AB - BA$:

$$C^T = (AB - BA)^T = (AB)^T - (BA)^T = B^T A^T - A^T B^T = BA - AB = -C$$

Since $C^T = -C$, it is a Skew-symmetric matrix.

Correct Option: (a)

7. **Value of α for $A + A^T = I$:**

$$\begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix} + \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \implies \begin{bmatrix} 2 \cos \alpha & 0 \\ 0 & 2 \cos \alpha \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$
$$2 \cos \alpha = 1 \implies \cos \alpha = \frac{1}{2} \implies \alpha = \frac{\pi}{3}$$

Correct Option: (b)

8. **Area of triangle calculation:**

$$\text{Area} = \pm 35 = \frac{1}{2} \begin{vmatrix} 2 & -6 & 1 \\ 5 & 4 & 1 \\ k & 4 & 1 \end{vmatrix} \implies 2(0) - (-6)(5 - k) + 1(20 - 4k) = \pm 70$$
$$30 - 6k + 20 - 4k = \pm 70 \implies 50 - 10k = \pm 70$$

Case 1: $50 - 10k = 70 \implies k = -2$

Case 2: $50 - 10k = -70 \implies k = 12$

Correct Option: (c)

9. **Finding $|3A|$ for a 3×3 matrix:**

Using property $|kA| = k^n |A|$ where $n = 3$:

$$|3A| = 3^3 |A| = 27|A|$$

Correct Option: (c)

10. **Matrix Inverse A^{-1} for $A = \begin{bmatrix} 2 & 3 \\ 5 & -2 \end{bmatrix}$:**

$$|A| = (2)(-2) - (3)(5) = -19$$

$$\text{adj } A = \begin{bmatrix} -2 & -3 \\ -5 & 2 \end{bmatrix} = -A \implies A^{-1} = \frac{1}{-19}(-A) = \frac{1}{19}A$$

Correct Option: (a)

11. **Simplify $(I + A)^3 - 7A$ when $A^2 = A$:**

$$(I + A)^3 = I^3 + 3I^2A + 3IA^2 + A^3 = I + 3A + 3A + A^2(A)$$

Since $A^2 = A$, this reduces to $I + 6A + A = I + 7A$.

Subtracting $7A$: $(I + 7A) - 7A = I$.

Correct Option: (c)

12. **Correct inverse property:**

By reversal law of matrix algebra: $(AB)^{-1} = B^{-1}A^{-1}$.

Correct Option: (b)

13. **Solving matrix equality:**

$x + y = 3$ and $x - y = 7$. Adding both gives $2x = 10 \implies x = 5$.

Then $5 + y = 3 \implies y = -2$.

Correct Option: (b)

14. **Possible number of orders for 24 elements:**

Find the integer pairs such that $m \times n = 24$:

$(1, 24), (24, 1), (2, 12), (12, 2), (3, 8), (8, 3), (4, 6), (6, 4)$. Total = 8 options.

Correct Option: (c)

15. **Determinant of odd-ordered Skew-symmetric matrix:**

The determinant of any 3×3 skew-symmetric matrix is always 0.

Correct Option: (a)

16. **Solve determinant value equality:**

$$\begin{vmatrix} x & 2 \\ 18 & x \end{vmatrix} = \begin{vmatrix} 6 & 2 \\ 18 & 6 \end{vmatrix} \implies x^2 - 36 = 36 - 36 \implies x^2 = 36 \implies x = \pm 6$$

Correct Option: (b)

17. **Find A^4 for $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$:**

$$A^2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I \implies A^4 = (A^2)^2 = I^2 = I$$

Correct Option: (b)

18. **Find $|\text{adj}(2A)|$ if $|A| = k$:**

First, $|2A| = 2^3|A| = 8k$. Then:

$$|\text{adj}(2A)| = |2A|^{3-1} = (8k)^2 = 64k^2$$

Correct Option: (c)

19. **If $AB = O$:**

Matrix multiplication permits divisors of zero, meaning neither A nor B must strictly be a zero matrix.

Correct Option: (c)

20. **Find λ such that $A^2 = I$:**

$$A^2 = \begin{bmatrix} \lambda & 1 \\ -1 & -\lambda \end{bmatrix} \begin{bmatrix} \lambda & 1 \\ -1 & -\lambda \end{bmatrix} = \begin{bmatrix} \lambda^2 - 1 & 0 \\ 0 & \lambda^2 - 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \implies \lambda^2 - 1 = 1 \implies \lambda = \pm\sqrt{2}$$

Correct Option: (d)

21. **Inverse Entry Property:**

Since $(A^{-1})_{ij} = \frac{C_{ji}}{|A|}$, entry $(A^{-1})_{23}$ matches up to the transposed position cofactor C_{32} .

Correct Option: (b)

22. **Find A^5 for $A = 2I$:**

$$A^5 = (2I)^5 = 32I = 16(2I) = 16A$$

Correct Option: (c)

23. **Find a, b if $(A + B)^2 = A^2 + B^2$:**

This condition implies $AB = -BA$. Calculating products:

$$AB = \begin{bmatrix} a-b & 2 \\ 2a-b & 3 \end{bmatrix}, \quad -BA = \begin{bmatrix} -a-2 & a+1 \\ -b+2 & b-1 \end{bmatrix}$$

Equating elements: $2 = a + 1 \implies a = 1$ and $3 = b - 1 \implies b = 4$.

Correct Option: (a)

24. **Collinear points calculation:**

$$\begin{vmatrix} x & -2 & 1 \\ 5 & 2 & 1 \\ 8 & 8 & 1 \end{vmatrix} = 0 \implies x(2-8) - (-2)(5-8) + 1(40-16) = 0$$

$$-6x - 6 + 24 = 0 \implies 6x = 18 \implies x = 3$$

Correct Option: (a)

25. **Determinant of Orthogonal Matrix:**

$$AA^T = I \implies |A|^2 = 1 \implies |A| = \pm 1.$$

Correct Option: (b)

26. **Find α for $A^2 = B$:**

$$A^2 = \begin{bmatrix} \alpha^2 & 0 \\ \alpha+1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 5 & 1 \end{bmatrix} \implies \alpha^2 = 1 \ (\alpha = \pm 1) \text{ and } \alpha + 1 = 5 \ (\alpha = 4)$$

No single parameter value can fulfill both equations.

Correct Option: (d)

27. **Find $|A|$ if $A^T = A^{-1}$:**

$$A^T A = I \implies |A|^2 = 1 \implies |A| = \pm 1.$$

Correct Option: (a)

28. **Evaluate $f(100)$:**

Factoring rows and columns systematically shows columns become linearly dependent, making the entire function evaluating cleanly to 0 for all real values of x .

Correct Option: (b)

29. **Row and Column matrix intersection order:**

To comply with both $1 \times n$ and $m \times 1$ shapes simultaneously, it must be 1×1 .

Correct Option: (a)

30. **Calculate $A^2 - 5A + 7I$:**

Using the Cayley-Hamilton Theorem for a 2×2 matrix: $A^2 - \text{tr}(A)A + |A|I = O$. Here $\text{tr}(A) = 5$ and $|A| = 7$, so it equates to O .

Correct Option: (b)

SECTION B: INTEGER TYPE QUESTIONS

31. **Value of $|A^{40}|$:**

$|A| = i^2 - 0 = -1$. Therefore, $|A^{40}| = |A|^{40} = (-1)^{40} = 1$.

Final Answer: 1

32. **Value of $|\text{adj}(\text{adj } A)|$:**

Formula: $|A|^{(n-1)^2}$. For $n = 3$ and $|A| = 2$:

$$2^{(3-1)^2} = 2^{2^2} = 2^4 = 16$$

Final Answer: 16

33. **Find non-negative integer x for singularity:**

$$(x+1)(x-2) - 6 = 0 \implies x^2 - x - 8 = 0$$

The roots formula outputs non-integer values ($x = \frac{1 \pm \sqrt{33}}{2}$); thus, 0 integer options exist.

Final Answer: 0

34. **Compute positive integer value of x :**

$$\begin{bmatrix} 2x & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -3 & 0 \end{bmatrix} \begin{bmatrix} x \\ 3 \end{bmatrix} = [0] \implies \begin{bmatrix} 2x-9 & 4x \end{bmatrix} \begin{bmatrix} x \\ 3 \end{bmatrix} = 2x^2 - 9x + 12x = 0$$

$$2x^2 + 3x = 0 \implies x(2x+3) = 0$$

No positive integer results.

Final Answer: 0

35. **Find $2 \cdot \det(2 \cdot A^{-1})$:**

$$\det(2 \cdot A^{-1}) = 2^3 \cdot \frac{1}{\det(A)} = \frac{8}{4} = 2 \implies 2 \cdot 2 = 4$$

Final Answer: 4

36. **Find λ for infinite solutions:**

Comparing plane equations $x + 2y + 3z = 10$ and $x + 2y + \lambda z = 10$, for dependency to exist, the coefficient positions must mirror perfectly. Hence $\lambda = 3$.

Final Answer: 3

37. **Find k for A^{10} :**

Computing powers shows a sequential pattern: $A^n = \begin{bmatrix} 1 & 0 \\ n & 1 \end{bmatrix}$. For $n = 10$, $k = 10$.

Final Answer: 10

38. **Maximum value of determinant:**

Row reductions transform the determinant cleanly down to structural form xy . Choosing maximum constraints from set $\{1, 2, 3\}$, we pick $x = 3, y = 3 \implies 3 \times 3 = 9$.

Final Answer: 9

39. **Find $|A|$ if $A^2 = 2A$:**

Taking determinant on both sides for order 3:

$$|A|^2 = 2^3|A| \implies |A|^2 = 8|A|$$

Since $|A| \neq 0$, dividing across by $|A|$ gives $|A| = 8$.

Final Answer: 8

40. **Find k for infinite solutions:**

Setting up coefficient proportionalities: $\frac{2}{4} = \frac{3}{k} \implies \frac{1}{2} = \frac{3}{k} \implies k = 6$.

Final Answer: 6